Soumitri Chattopadhyay

UC San Diego, CA, USA

https://soumitri2001.github.io | sochattopadhyay@ucsd.edu

Google Scholar | GitHub | LinkedIn

Research Interests

Methods: Foundational Models, Multi-modal learning (Vision+X), In-Context learning, Visual Agentic AI
Applications: Medical Imaging, Longitudinal Analysis, Clinical Workflows; Image Restoration

Education

University of California, San Diego — California, USA
Ph.D. in Computer Science and Engineering — *Sept. 2025 – May 2028 (Expected)*

University of North Carolina at Chapel Hill — North Carolina, USA
M.S. in Computer Science (Ph.D. Transferred to UCSD) — *Aug. 2023 – May 2025*

Jadavpur University — Kolkata, India
B.E. in Information Technology | CGPA (out of 10): 9.01 — *Aug. 2019 – June 2023*

Research Experiences

UC San Diego — California, USA
Graduate Student Researcher — *Sept. 2025 – Present*

- Part of the UCSD Biomedical Image Analysis Group, advised by Prof. Marc Niethammer.
- Currently working on in-context promptable foundational models for multimodal 2D/3D medical segmentation.
- Devised cross-modal visual prompting approaches for 3D medical segmentation with foundational models.
- Analyzed robustness of segmentation foundational models against imprecise visual prompts **(ISBI'26 Oral)**.
- Developed a visual agentic workflow to process, segment and extract radiomics features from 2D and 3D raw clinical imaging data for early diagnosis of Lupus (as a part of an **NIH grant**).

UNC Chapel Hill — North Carolina, USA
Graduate Research Assistant — *Aug. 2023 – Sept. 2025*

- Part of the erstwhile UNC-BIAG, advised by Prof. Marc Niethammer. Transferred to UCSD upon advisor's move.
- Conducted an extensive study on zero-shot domain generalizability of medical segmentation foundational models.
- Proposed a few-shot latent diffusion model-aided fine-grained image recognition **(BMVC '24 Oral)**.

SRI International — Menlo Park, CA
Research Intern — *May 2024 – Dec. 2024*

- Interned in the NuSCI Research Group, supervised by Dr. Anirban Roy
- Worked on **brain decoding**; developed an end-to-end framework for visual decoding from fMRI scans with controllable latent diffusion models and various training recipes that outperformed prior SOTA works.

École de technologie supérieure (ÉTS), Montreal — Montreal, Canada
MITACS Globalink Research Intern — *June 2022 – Aug. 2022*

- Advisor: Prof. Jose Dolz
- Explored handling of 3D medical volumes (CT and MRI), including pre-processing, sampling, and visualization.
- Implemented a knowledge distillation-aided baseline for unpaired multi-modal medical image segmentation.

CVPR Unit, Indian Statistical Institute — Kolkata, India
Research Intern — *Nov. 2021 – May 2023*

- Supervisors: Prof. Umapada Pal and Prof. Saumik Bhattacharya
- Investigated the role of temperature in Self-supervised Contrastive loss and proposed a dynamic temperature scaling to improve its performance **(IEEE TAI '24)**.
- Developed Self-Supervised Learning frameworks for Offline Signature Verification **(ICIP '22 Oral, ICPR '22)**.

Selected Publications / Preprints

(* denotes equal contribution)

15. D. Mandal, **S. Chattopadhyay**, Y. Wang, M. Niethammer, P. Chakravarthula, UnSCAR: Universal, Scalable, Controllable, and Adaptable Image Restoration, *Under Review.*
14. **S. Chattopadhyay**, B. Demir, M. Niethammer, Uncertainty-Aware Spatio-Semantic Contextual Prompts for Multimodal Medical Segmentation, *Under Review.*
13. **S. Chattopadhyay**, Y. Wang, D. Mandal, B. Demir, M. Niethammer, P. Chakravarthula, Diffractive Optical Neural Networks for Ultra–Light-Weight Medical Image Segmentation, *Under Review.*
12. B. Demir, **S. Chattopadhyay**, H. Greer, B. Chen, M. Niethammer, How Useful Are Vision Foundation Model Features for Out-of-the-Box Disease Progression Prediction?, **arXiv** (Under Review.)
11. **S. Chattopadhyay**, B. Demir, M. Niethammer, On the Robustness of Foundational 3D Medical Image Segmentation Models Against Imprecise Visual Prompts, **ISBI 2026, Oral**
10. D. Mandal, **S. Chattopadhyay**, G. Tong, P. Chakravarthula, UniCoRN: Latent Diffusion-based Unified Controllable Image Restoration Network across Multiple Degradations, **WACV 2026**, **Oral**
9. **S. Chattopadhyay**, S. Biswas, E. Vivoli, J. Llados, Towards Generative Class Prompt Learning for Fine-grained Visual Recognition, **BMVC 2024**, **Oral**
8. A. Sain, A.K. Bhunia, S. Koley, P.N. Chowdhury, **S. Chattopadhyay**, T. Xiang, Y.Z. Song, Exploiting Unlabelled Photos for Stronger Fine-Grained SBIR, **CVPR 2023**
7. H. Thakur* and **S. Chattopadhyay***, Active Learning for Fine-Grained Sketch-Based Image Retrieval, **BMVC 2023**
6. M. Bhattacharyya*, **S. Chattopadhyay***, S. Nag*, DeCAtt: Efficient Vision Transformers with Decorrelated Attention Heads, **CVPR 2023 – ECV Workshop**, **Oral**
5. **S. Chattopadhyay**, S. Ganguly, S. Chaudhury, S. Nag, S. Chattopadhyay, Exploring Self-Supervised Representation Learning For Low-Resource Medical Image Analysis, **ICIP 2023**
4. A. Deb*, S. Nag*, A. Mahapatra*, **S. Chattopadhyay***, A. Marik*, *et al.* BeAts: Bengali Speech Acts Recognition using Multimodal Attention Fusion, **INTERSPEECH 2023**, **Oral**
3. H. Basak, **S. Chattopadhyay**, R. Kundu, S. Nag, R. Mallipeddi, IDEAL: Improved DEnse locAL Contrastive Learning for Semi-Supervised Medical Image Segmentation, **ICASSP 2023**
2. S. Manna, **S. Chattopadhyay**, S. Bhattacharya, U. Pal, SWIS: Self-Supervised Representation Learning For Writer Independent Offline Signature Verification, **ICIP 2022**, **Oral**
1. **S. Chattopadhyay**, S. Manna, S. Bhattacharya, U. Pal, SURDS: Self-Supervised Attention-guided Reconstruction and Dual Triplet Loss for Offline Signature Verification, **ICPR 2022**

Technical Skills

Frameworks/Libraries: PyTorch, TensorFlow, MONAI, PyRadiomics, NumPy, Pandas, Matplotlib
Softwares/Developer Tools: LaTeX, VSCode, Slicer, ITK-SNAP, Android Studio

Academic Services

Peer-reviewing (Since 2022)

- **Conferences:** CVPR, MICCAI, WACV, ISBI, ICCV, ACM MM, ICPR; ICCV + CVPR Workshops
- **Journals:** International Journal of Intelligent Systems, Wiley **;** Engineering Applications of Artificial Intelligence, Elsevier **;** Computers in Biology and Medicine, Elsevier **;** IEEE Access